

Two cycles

X17 (2017) — English translation

The diagram in coordinates (ρ, T) shows two cycles (see figure), according to which heat engines I and II operate. They use helium as a working fluid. These cyclic processes have efficiency factors η_1 and η_2 , satisfying the condition: $(3 - \eta_1)(1 - \eta_2) = 1$.

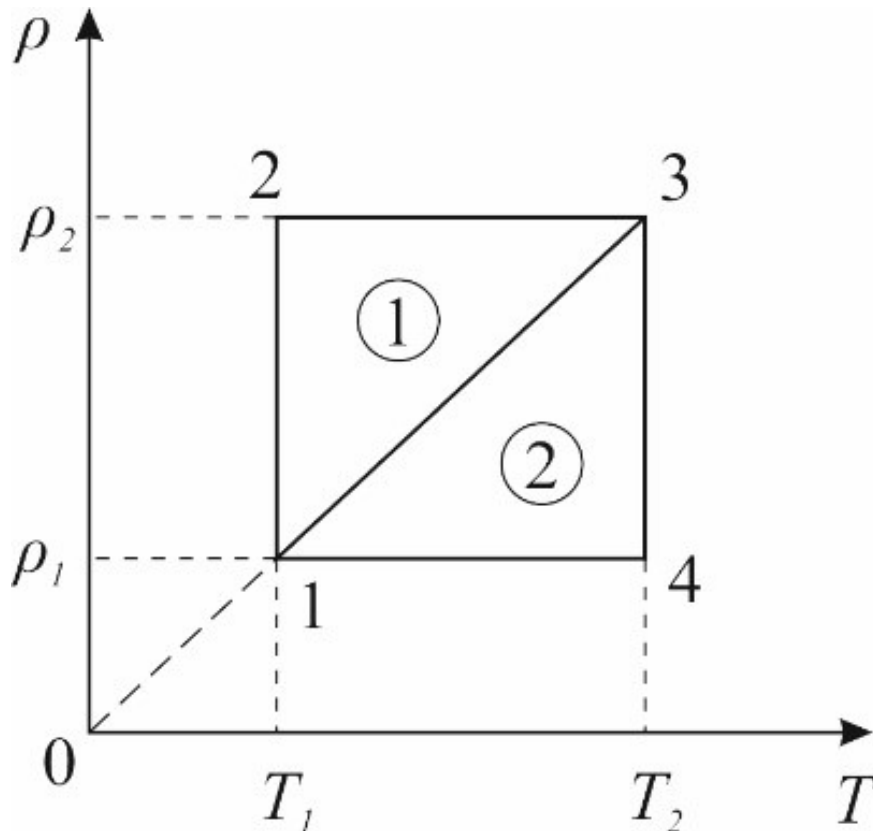


Figure 1: Figure 1.

A1	Draw (qualitatively) this diagram in coordinates (P, V) .	2.00
A2	Calculate temperature T_2 if temperature $T_1 = 50$ K.	8.00

Source: <https://pho.rs/p/306>